DEEPTANSHU DEVATHA

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EDUCATION

Heritage High School, Frisco, TX — Class of 2027

GPA 4.00 UW / 5.361 W · Rank **11/486** · ACT **36** · AP Scholar with Distinction

AP exams: 9 taken, a 5 on all 9 — Computer Science A, Statistics, Physics 1, Seminar, English Language & Composition, U.S. History, World History, Psychology, Human Geography

Coursework: AP Calculus BC, AP Physics 2, AP Physics 1, AP Computer Science A, AP English Language, AP English Literature, AP Government, AP Economics, Computer Science III, Mobile App Programming, Spanish 2

HONORS & AWARDS

IEEE-EMBS IECBES 2026 — Research Paper Accepted for Presentation — Dec. 9–11, 2026

Conrad Innovator 2025–2026 — Health & Nutrition · X-Step startup — 2025–2026

Texas A&M McFerrin Center for Entrepreneurship Texas Ideas Challenge — Top 40 Finalist · X-Step — May 2026

Diamond Challenge 2026 — Semifinalist · X-Step startup — 2026

TechLit Bridging the Gap Hackathon — 1st Place · ShopAI (as ShopSmart) — Aug. 2025

PeddieHacks — Best Innovation Award (1st/250) · ShopAI (as ShopSmart) — Aug. 2025

Hack4Humanity — 3rd Place (3/500) · ShopAI (as ShopSmart) — Oct. 2025

AsterHacks — 5th Place (5/250) · ShopAI (as ShopSmart) — Oct. 2025

Competitions: deeptanshudevatha.com/projects/competitions ·
Texas A&M list: mcferrin.tamu.edu/program/texas-high-school-ideas-challenge ·
TechLit submission: devpost.com/software/smartshop-gj4q59 · **IECBES paper:** deeptanshudevatha.com/research/x-step

RESEARCH

PAPER 1 · COMPUTATIONAL LINGUISTICS / CLINICAL ML

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features — May 2025–Present

First Author · Mentor: Joe Xiao (PhD student)

medRxiv preprint (Aug. 2026) · **Revision recommended by the editors, Frontiers in Aging Neuroscience — submitted August 04, 2026**

Preprint: medrxiv.org/content/10.64898/2026.08.04.26359730v1 · **Code:** github.com/ddevatha1/dementia-mmse-prediction ·
Write-up: deeptanshudevatha.com/research/speech-cognition

Submission post: deeptanshudevatha.com/#modal-dementia-25 · **leakage fix:** deeptanshudevatha.com/#modal-dementia-12 ·
data-join bugs: deeptanshudevatha.com/#modal-dementia-14 · **ablation:** deeptanshudevatha.com/#modal-dementia-24

QUESTION — Can transcribed speech plus the clinical history already on file forecast a patient's **next-visit MMSE** (0–30)? Framed as longitudinal, patient-level supervised regression; most prior speech work is single-visit classification.

DATA — DementiaBank Pitt Corpus (controlled access): **126 patients** (69 dementia, 57 control), 2–5 visits → **173 samples**.

FEATURES — Designed a 53-dimensional feature vector: 12 per-visit features — Sentence-BERT discourse coherence (adjacent-sentence similarity, global coherence, semantic slope, topic drift) plus spaCy lexical/syntactic measures (filler rate, 50-token MATTR, inter-sentence repetition, POS-based fragment ratio, named-entity rate) — with 8 visit-to-visit Δ features, 4 temporal features, 20 PCA components of 384-d SBERT embeddings, and 9 clinical/demographic covariates.

MODEL — LightGBM, chosen for small mixed-type tabular data: RobustScaler → correlation pruning ($|r| > 0.8$, clinical variables kept) → permutation-importance selection (top 15) → 3× Gaussian augmentation tagged by parent patient → Optuna TPE over 10 hyperparameters, 200 trials, maximizing mean patient-grouped R^2.

VALIDATION — Diagnosed patient-identity leakage: folds split by visit, so a model could score by recognizing a person. Rebuilt evaluation as **5-fold GroupKFold on patient ID**, locking augmented copies to their parent's fold and scoring only **out-of-fold predictions on original samples**; scores fell once leaked identity was removed.

DEBUGGING — Found why entry MMSE, the expected top predictor, contributed almost nothing: two silent data-join bugs (renamed column; string-vs-integer patient ID) imputed one value for every patient. Fixing them was the largest single gain.

RESEARCH (continued)

RESULTS R^2 **0.840 ± 0.015** · **RMSE 2.75** · **MAE 2.02** MMSE points · **Pearson r 0.917**. Entry MMSE, baseline diagnosis and Blessed score lead permutation importance.

ABLATION Controlled feature-source ablation: linguistic-only R^2 **0.118**, SBERT-only **0.458**, clinical-only **0.716**, clinical + SBERT **0.759**. Clinical history dominates; SBERT beats handcrafted features and adds complementary signal.

INTEGRITY & LIMITS Audited the manuscript against the code, removing six unused features. Limits: non-nested selection and tuning (CV may be optimistic); one cohort, no external validation; over-prediction below MMSE 10.

PAPER 2 · BIOMEDICAL ENGINEERING / WEARABLE SENSING

A Four-Site Plantar-Pressure Insole: Transducer Error Budget, Feature Equivariance, and Sparse-Versus-Dense Agreement in Overground Walking

Jul. 2026–Present

Accepted for Presentation · 2026 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES) · Kuala Lumpur · Dec. 9–11, 2026

2nd of 5 authors · ML lead · UC Santa Cruz AIEA Lab (Prof. Leilani Gilpin)

Paper page: deeptanshudevatha.com/research/x-step · **Lab:** aiea-lab.github.io

QUESTION Can **four strategically placed sensing sites** (MET1, MET2, MET5, heel; ESP32, 25 Hz) capture regional pressure trends comparable to a **32-sensor insole**? I engineered the machine-learning component and designed the correlation study that tests it, working from the pressure measurements our team's four-site system generates.

ML PIPELINE Built the analysis pipeline for this question: the feature map over rolling 4 s windows (100 samples × 8 channels) to **59 features** — per-site peak, mean, coefficient of variation and pressure–time integral; window-level cadence, stance ratio, anterior load share, aggregate peak and total PTI — and a **540-parameter** logistic-regression framework that classifies biomechanical risk from them (≈0.23 ms per update).

CORRELATION STUDY Designed the sparse-vs-dense study against an independent 32-cell insole archive (**149 recordings**, **15 adults**, 298 foot-takes), comparing the cell nearest each of the insole's four sites with its anatomical cluster. Shape survived (**median r 0.72–0.99**), magnitude did not (**peak-to-peak r 0.43–0.74**); four-site center of pressure tracked anteroposterior (**r 0.81**) but not mediolateral motion (**r 0.01**) — a limit of the four-site geometry. Descriptive only.

EVALUATION Quantified subject-identity leakage on a synthetic 24-subject, 9-class cohort: **macro-F1 0.885** subject-grouped vs **0.931** ungrouped (majority baseline 0.040) — a gap larger than the spread across classifiers.

PROJECTS & LEADERSHIP

ShopAI — *Creator & Sole Developer*

2025–Present

One grocery search across nine retailers, plus multi-store trip planning · live on the Apple App Store

New name for ShopSmart, the team hackathon prototype; ShopAI extends its core idea as a solo rebuild. Won TechLit (1st), PeddieHacks (Best Innovation), Hack4Humanity (3rd) and AsterHacks (5th) as ShopSmart.

App Store: apps.apple.com/app/shopai/id6797083809 · **Project page:** deeptanshudevatha.com/projects/shopai · **Architecture:** deeptanshudevatha.com/#modal-shopai-architecture · **Hackathon page:** devpost.com/software/smartshop-gj4q59

ARCHITECTURE React Native/Expo client over a standalone Express/TypeScript backend on **AWS EC2 behind Nginx**; the client holds no search logic — every search, plan and warm-up check runs server-side through a four-layer, one-direction pipeline.

ADAPTERS Designed a **retailer-adapter abstraction** normalizing **nine chains** (seven with live data) into one schema via four integration strategies: OAuth2 client-credentials REST (Kroger, Harris Teeter), anonymous-session GraphQL (Aldi, Sprouts, Publix via Instacart), headless Playwright (Trader Joe's), server-rendered HTML (Whole Foods). Target excluded (terms of service).

SEARCH PATH Query correction → **concurrent fan-out** racing a shared response-time budget → per-store food/relevance filtering and ranking → explicit ok / error / unavailable / pending status per store. A per-ZIP **state machine** (starting → warming → ready) pre-warms OAuth tokens, cookies and the browser session off the request path.

PLANNER A **deterministic optimizer** exhaustively scores all $2^7 - 1 =$ **127** non-empty store subsets, so "cheapest" is exact; cheapest, fastest and fewest-stops are single-field sorts; only "balanced" is weighted multi-criteria. **Zero** LLM calls in search or planning.

PROJECTS & LEADERSHIP (continued)

PRODUCTION Debugged a Node.js setTimeout overflow in a 30-day location cache that stalled the event loop (≈1.5 s → 27 s searches) by capping the sweep interval; shipped by feature-flagging off unfinished meal planning, AI assistant, OCR and barcode scanning.

KidCode — *Founder & Lead Instructor* · Frisco, TX — Jun. 2024–Present

124 students · 3-year program · 10 Frisco neighborhoods · **120 one-hour classes** · **20 projects** · median 10 per student (max 18)

Program: deeptanshudevatha.com/projects/kidcode · **Curriculum:** deeptanshudevatha.com/projects/kidcode/curriculum ·
Assessment: deeptanshudevatha.com/projects/kidcode/assessment · **Student programs:** deeptanshudevatha.com/projects/kidcode/projects

CURRICULUM Designed a three-year curriculum of 120 one-hour classes (8 weeks each summer, 5 per week) and 20 projects (6 + 7 + 7): Year 1 Python foundations; Year 2 assert-based unit tests, file I/O, JSON, exception handling, classes and inheritance, libraries; Year 3 spaCy NLP — tokenization, lemmatization, sentence segmentation, POS tagging, NER, Matcher rule patterns, and a rule-based chatbot.

SPIRAL DESIGN Earlier projects return at a higher level: Mad Libs becomes Auto Mad Libs with spaCy choosing the blanks; the trivia quiz returns JSON-backed with error handling; Year 2 ends by breaking students' own text-cleaning code, which Year 3 opens by testing against spaCy.

ASSESSMENT Wrote 13 curriculum-aligned pre/post assessment pairs (same skills and formats, different code). Paired results, **n = 104**: mean **41.7% → 58.8%** (**+17.1 pp**); **99/104** improved, 4 unchanged, 1 declined. Observed change only; no comparison group.

STUDENT WORK Students completed **1,214** working programs across the 20 projects, from Python basics in Year 1 to spaCy NLP in Year 3 (all **572** Year 3 programs use spaCy).

X-STEP STARTUP · PRODUCT BUILT ON IEEE-EMBS IECBES 2026 ACCEPTED RESEARCH

X-Step — *Co-Founder & ML/Mobile Lead* — Aug. 2025–Present

Student startup team · working prototype, no wearer data yet · Conrad Innovator 2025–2026, Texas A&M Texas Ideas Challenge Top 40 Finalist, Diamond Challenge 2026 Semifinalist

Project page: deeptanshudevatha.com/projects/x-step · **Paper page:** deeptanshudevatha.com/research/x-step

PRODUCT A plantar-pressure insole for people at risk of diabetic foot problems, built by the team: four FSR402 sensors per foot on an ESP32 at 25 Hz, streaming sequence-numbered 28-byte BLE frames to a React Native (Expo) app with live pressure, trends, alerts and a clinician view.

MY ROLE Wrote the React Native/Expo app (BLE client, live pressure heatmap, alerts, trends, guided routine; earlier Flask + LLM health-assistant backend) and trained two CNN prototypes on public data (heatmap postures, wound severity). Designed the ML pipeline that turns the stream into alerts: rolling 4 s windows → 59 pressure and gait features → 540-parameter logistic-regression head, fast enough to run between 40 ms samples.

Ayusha Nilayam — *Programming & Math Instructor* · Hyderabad, India — Nov. 2025–Jul. 2026

Project page: deeptanshudevatha.com/projects/ayusha-nilayam · **Zoom sessions post:** deeptanshudevatha.com/#modal-india-01

SCOPE Taught **38** weekly Zoom sessions of Python and math to **11 children** at the Ayusha Nilayam orphanage in the Alwal/Kompally area of Hyderabad, plus two weeks in person.

CONTENT Python from print() through conditionals, loops and functions to independent projects (calculator, number-guessing game); math from equations and inequalities through the unit circle, and statistics through standard deviation.

CONSTRAINTS Adapted to one shared laptop, unreliable internet and a Telugu–English language gap with frequent checks for understanding and student explain-backs.

UT Dallas NLP with Python Workshop — *Participant* — Summer 2025

Dr. Chris Davis · Rule-based text simplifier: dependency parsing, lexical substitution, clause splitting, passive-to-active rewriting.

TECHNICAL SKILLS

ML & NLP LightGBM, scikit-learn, Optuna (TPE), PyTorch, PCA, permutation importance, GroupKFold, ablation design; spaCy (tokenization, POS, NER, Matcher, dependency parsing), Sentence-BERT; windowed sensor-feature extraction

LANGUAGES Python, Java, TypeScript/JavaScript, SQL

TECHNOLOGIES Mobile and backend: React Native/Expo, Node.js/Express, REST/GraphQL, OAuth2, Playwright; AWS EC2, Nginx